SQL Query AI Agent

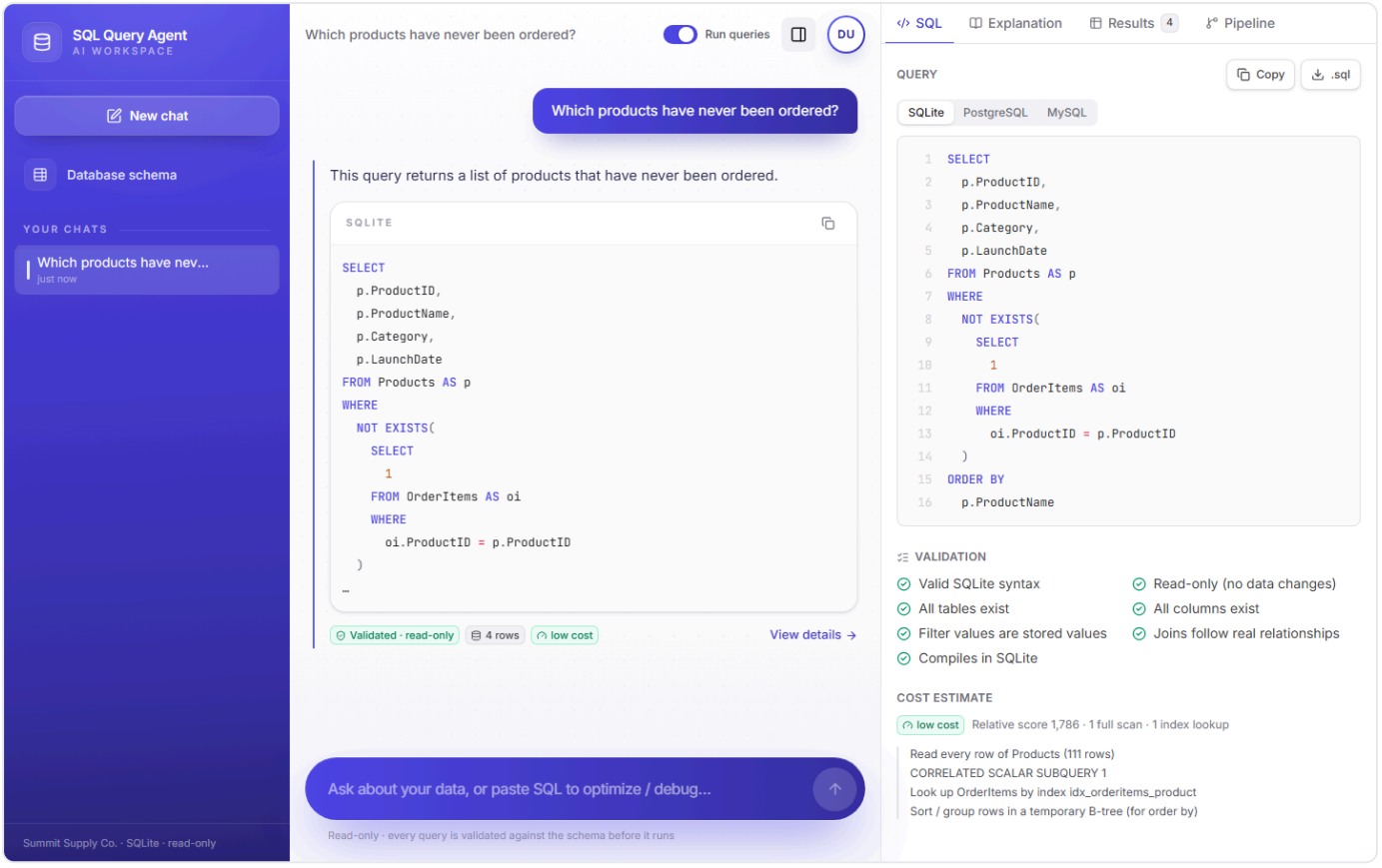
An AI assistant that turns plain-English questions into **validated, read-only SQL**. It also explains, optimizes and debugs queries, holds a multi-turn conversation, and refuses anything outside its scope. It is built as a **LangGraph** state machine with a **FastAPI** backend, a **React** chat UI, and **Claude Haiku 5.5 on AWS Bedrock**.

> **Live demo:** *link added after deployment*

Test logins

Account	Username	Password	What it shows
Normal user	`demo`	`demo-sql-2026`	The end-user experience: chat, SQL, explanation, results, CSV, cost, validation, pipeline summary
Developer	`developer`	`dev-sql-2026`	Everything above, plus the **Dev** and **Runs** tabs (trace, every LLM call with tokens, cost and time, run history), the prompt library and the knowledge graph

Both logins are defined in `backend/.env` (`AUTH_NORMAL_*` , `AUTH_DEV_*`) and can be changed there.



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What it does

You ask	The agent
"Show all employees hired after January 2024."	Writes the SQL, validates it, runs it (135 rows) and explains it in plain English
"Show all customers." → "Only those from California."	Treats the second message as a follow-up and **edits the previous query** (`WHERE c.State = 'CA'` , 343 rows)
"Which sales reps generated the most net revenue from West Coast customers in 2024? Top 5."	Plans a 4-table join, applies the business definition of *net revenue*, maps *West Coast* to `IN ('CA','OR','WA')`
"Optimize this query: `SELECT * FROM Orders WHERE …` "	Rewrites it and accepts the rewrite **only if it returns identical rows**, with cost before/after and index advice
"Why is this query failing? `SELECT … FROM Customer WHERE State = 'California'` "	Lists each issue and why it is wrong, then returns a corrected query that has been validated and executed
"Explain this query: …"	Gives a step-by-step plain-English explanation
"What does the Segment column mean?"	Answers from the knowledge base documentation
"Who won the FIFA World Cup?"	*"I'm designed to assist only with SQL and database-related tasks. Please ask a question related to the provided database schema."*
"Delete all cancelled orders"	Refuses, and offers a read-only `SELECT` that shows the affected rows instead

Every answer comes with the SQL (with syntax highlighting, copy and download), the explanation, the results (with a full CSV export), a validation checklist, a cost estimate, PostgreSQL/MySQL versions of the query, and three grounded **"Dive deeper"** follow-up questions.

Assignment coverage

Core requirements

Requirement	Status	Where
Natural language → SQL	✅	`query_planner` → `plan_grounding` → `sql_generation`
SQL validation (syntax, tables, columns, read-only)	✅	Deterministic `SQLValidator` (sqlglot AST + SQLite compile)
Block DELETE / UPDATE / INSERT / DROP / ALTER / TRUNCATE	✅	Four independent layers, see Safety
Query explanation	✅	`generate_explanation` (streamed)
Query optimization	✅	`sql_optimization` , with a result-equivalence check
Query debugging	✅	`sql_debugging` + validator findings
Conversation context / follow-ups	✅	LangGraph checkpointer, previous plan and SQL are edited
Out-of-scope refusal (exact wording)	✅	`scope_validation`
LangGraph orchestration	✅	17-node `StateGraph` with a repair loop, parallel fan-out and memory
Chat UI, SQL panel, explanation panel, history, loading state, errors, copy	✅	React app

Bonus features

Bonus	Status	Notes
Query execution against a sample DB	✅	Read-only SQLite, 5 s timeout, 500-row preview + true total count
Follow-up suggestions	✅	"Dive deeper": drill-down / refine / related, checked against the schema
Query cost estimation	✅	From SQLite's `EXPLAIN QUERY PLAN`: full scans, index use, temp sorts, index DDL advice
SQL syntax highlighting	✅	Custom highlighter, no heavy editor dependency
Multiple SQL dialects	✅	SQLite (executed) plus PostgreSQL and MySQL versions via sqlglot
CSV export / download SQL	✅	Server-side CSV re-runs the validated SQL (up to 50,000 rows); `.sql` download
Prompt injection protection	✅	Deterministic input guard **before any LLM call**, plus prompt-level rules and output validation
Dark mode, streaming responses	✅	Theme toggle; SSE progress per node + token-streamed explanation
Unit tests	✅	137 tests, no network needed (scripted LLM)
Docker / deployment	⏳	Next step

Deliverables

Deliverable	Where
Source code	`backend/`, `frontend/`
README	this file
Architecture diagram	docs/architecture.md
LangGraph workflow explanation	docs/workflow.md
Prompts used	docs/prompts.md (generated from the code)
Sample database schema	docs/database.md and backend/data/schema.sql
Assumptions	docs/assumptions.md
PDF copies of every document	`docs/pdf/`
Deployed link	Linked at the top once available

Quick start

Prerequisites

- Python **3.11+** (developed on 3.14)
- Node.js **20+**
- An **AWS Bedrock API key** (bearer token) with access to Claude Haiku 5.5 in `us-west-2`

1. Backend

```
cd backend
python -m venv .venv
# Windows: .venv\Scripts\activate    macOS/Linux: source .venv/bin/activate
pip install -e ".[dev]"

cp .env.example .env
# Fill in AWS_BEARER_TOKEN_BEDROCK, the two test-account passwords and AUTH_SECRET

python scripts/check_llm.py          # confirms Bedrock answers on both model tiers
uvicorn main:app --reload --port 8000
```

The sample database (`data/company.db`) and the knowledge base (`knowledge_base/*.json`) are committed, so no build step is needed. To regenerate them:

```
pip install -e ".[data]"
python scripts/generate_data.py --xlsx   # deterministic synthetic data (+ Excel copy)
python scripts/build_knowledge.py        # data dictionary, knowledge graph, relations_graph.html
python scripts/generate_docs.py          # docs/prompts.md, docs/database.md, graph in docs/workflow.md

pip install -e ".[docs]" && python -m playwright install chromium
python scripts/export_docs_pdf.py        # README + docs/*.md → docs/pdf/*.pdf
```

2. Frontend

```
cd frontend
npm install
cp .env.example .env
npm run dev                          # http://localhost:5173 (proxies /api to :8000)
```

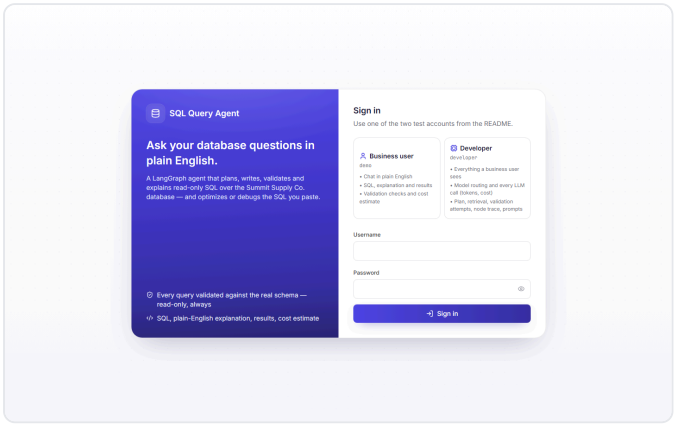
3. Single-process mode (as deployed)

```
cd frontend && npm run build         # writes frontend/dist
cd ../backend && uvicorn main:app --port 8000
# http://localhost:8000 serves the UI and the API
```

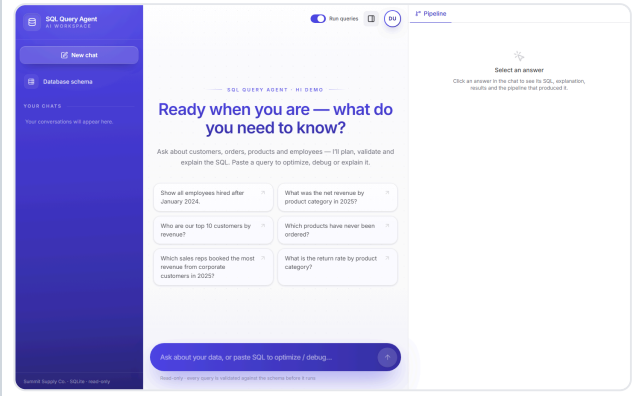
4. Terminal client (optional)

```
cd backend
python scripts/ask.py "Show all employees hired after January 2024."
python scripts/ask.py --thread demo "Show all customers." "Only those from California."
python scripts/ask.py --dev "Top 5 customers by revenue"     # also prints the trace and every LLM call
```

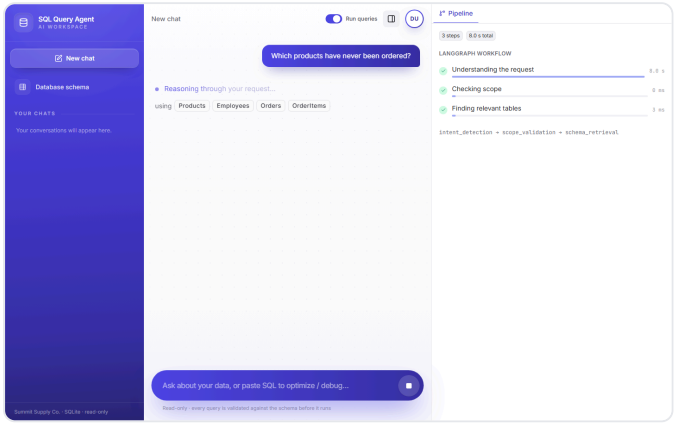
Using the app

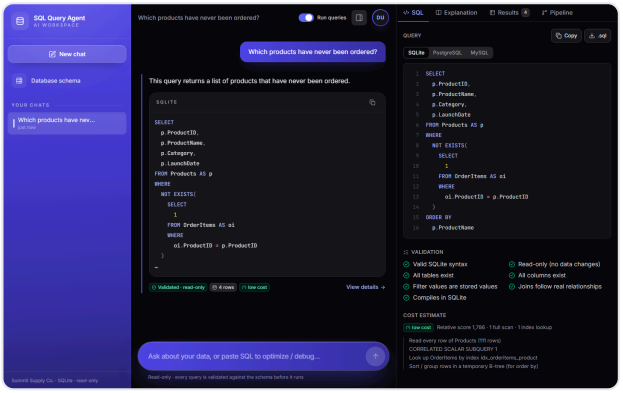


Login. Two test accounts, defined in `backend/.env`



Welcome. Example questions and a schema browser





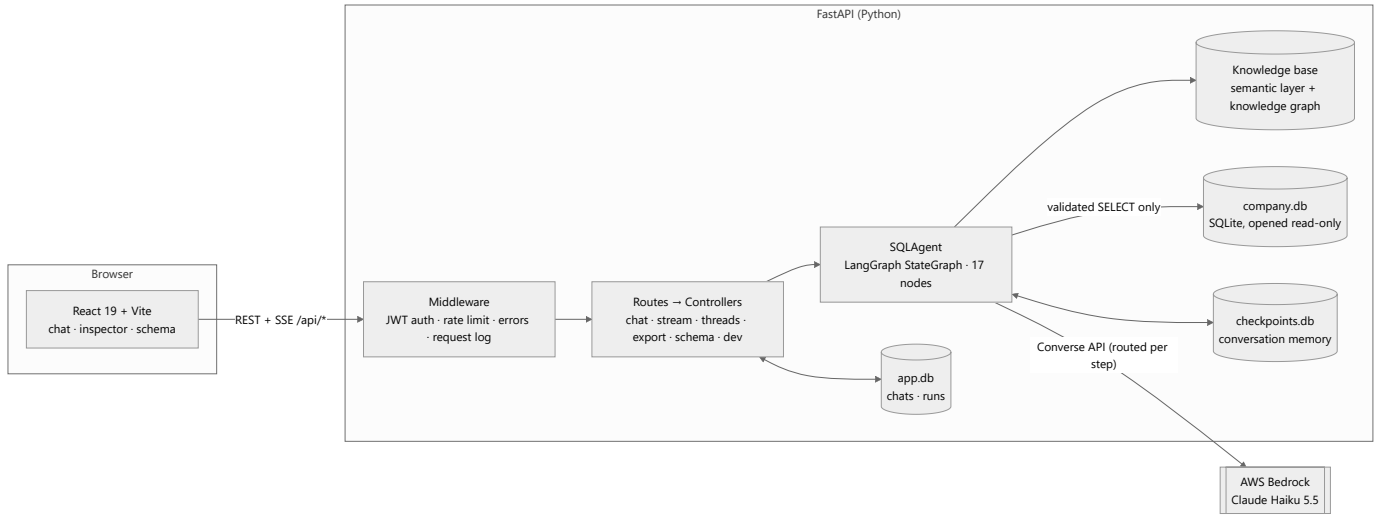
Streaming. A live caption follows the graph ("Planning the query…", "Validating the SQL…")	**Dark mode**

Two roles. Both accounts use the same agent. The role only changes what the API returns:

	`demo` (normal)	`developer`
Chat, SQL, explanation, results, CSV, cost, validation checklist, pipeline summary	✅	✅
Dev tab: intent, retrieval, plan, grounding, every validation attempt, node trace, every LLM call (model, routing reason, tokens, cost, latency), usage totals table	–	✅
Runs tab: history of past runs with tokens, cost and timing	–	✅
Prompt library and knowledge-graph viewer	–	✅

The role is enforced on the server (`/api/dev/*` returns 403 for the normal account, and the response's `dev` section is stripped). It is not just hidden in the UI.

Architecture at a glance



The guiding idea: **the LLM decides what to ask; deterministic code decides whether it is correct and safe.** Anything that can be checked without a model is: names, stored values, joins, read-only-ness, cost and result equivalence. The model only sees a schema slice that has been grounded against the real data, and every SQL it writes goes through the validator before it can reach the database.

More detail, including sequence diagrams, storage and security layers: **docs/architecture.md**.

The LangGraph pipeline

```
receive_request → intent_detection → scope_validation ─┬→ return_response    (refusal)
                                                       ├→ ask_clarification  (optimize/debug/explain without SQL)
                                                       ├→ schema_answer      (questions about the database)
                                                       └→ schema_retrieval
schema_retrieval ─┬→ query_planner → plan_grounding ─┬→ ask_clarification   (unknown field / value / ambiguity)
                  │                                  └→ sql_generation → sql_validation
                  └→ sql_validation                   (user-supplied SQL)

sql_validation ─┬─ valid ──→ [sql_debugging] → sql_optimization → execute_query ─┬→ generate_explanation ┐ parallel
                │                                                                └→ suggest_follow_ups   ┘
                ├─ invalid, user's SQL    → sql_debugging → sql_validation
                ├─ invalid, attempts left → sql_repair    → sql_validation       (self-correction, max 3)
                └─ invalid, budget spent  → return_response
every terminal node → return_response → END
```

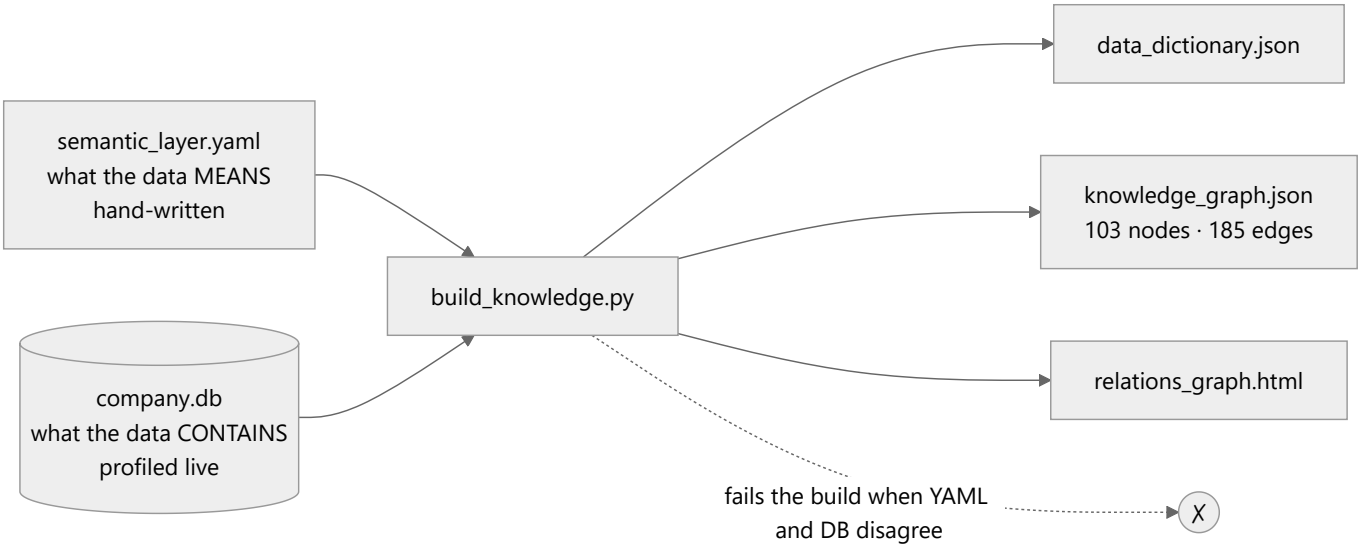
Feature	How
One file per node	`agent/<area>/<node_name>.py` , e.g. `exec/sql_generation.py`

Feature	How
All branching in one place	`supervisor/routing.py`
Self-correction	`sql_validation` ⇄ `sql_repair` cycle; the repair prompt gets each finding's code, message and fix hint
Parallel fan-out	`generate_explanation` ‖ `suggest_follow_ups` in one superstep; an append reducer merges their trace/LLM records
Memory	Checkpointer keyed by `thread_id` (SQLite in production). Follow-ups edit the **previous plan** and **previous SQL**
Streaming	`graph.stream(stream_mode=["updates","messages"])` → one SSE event per node + explanation tokens
Resilience	`traced()` wraps every node: timing, trace event, and any exception becomes a clean `failure`

Node-by-node description, the state schema, routing table and walkthroughs: **docs/workflow.md**.

Knowledge graph and grounding

Generating SQL from table names alone fails on real data. "California" is stored as `CA`, "revenue" excludes cancelled orders, and "sales rep" means `Orders.EmployeeID`. The agent therefore works from a knowledge base, built the same way as in the intelligence-assistant:



- **Graph:** 5 tables, 57 columns, 21 metrics and 20 glossary terms as nodes. Edges are `has_column`, `references`, `joins_to` and `uses`. It also holds every join path, validated metric SQL, and indexes of stored values and schema terms.
- **Retrieval swarm** (`retrieval/swarm.py`): several retrievers run in parallel over schema terms, stored values, metrics and glossary, similar worked examples (14 Q→SQL pairs in `examples.yaml`), tables in pasted SQL, and the previous turn's tables. FK bridge tables are added automatically.
- **Phrase matching** (`knowledge/kb_resolver.py`): it takes the longest non-overlapping match ("west coast" beats "west"). US state codes match only in upper case, so "in" is not Indiana. Table words and stored values are told apart ("products" the table vs the *Product* department).
- **Plan grounding** (`planner/plan_grounding.py`): the planner writes values *as the user said them*. Grounding snaps names to the real schema, maps values to stored values (`California` → `'CA'`, regions → `IN` lists, product names through a database `LIKE` search), attaches metric definitions, and computes the minimal FK join tree (an exact Steiner tree over the 5 tables). It also scores complexity. Anything it cannot resolve becomes a **clarifying question** instead of a guess.

Safety and guardrails

Write protection is layered, so no single failure can modify data:

Layer	Where	What it stops
1. Input guard	`guardrails/input_guard.py`, before any LLM call	Prompt injection patterns (override instructions, reveal prompt, role hijack, jailbreak modes, fake delimiters), pasted write SQL, empty or over-long input
2. Intent + scope	`intent_detection` → `scope_validation`	Destructive and out-of-scope requests are refused deterministically

Layer	Where	What it stops
3. SQL validator	`guardrails/sql_validator.py`	Anything that is not one read-only `SELECT`, plus unknown tables/columns/values, joins not backed by a foreign key, SQL SQLite cannot compile, and SQL that ignores the plan's filters
4. Read-only database	`exec/sqlite_runner.py`	`mode=ro` URI, `PRAGMA query_only`, an authorizer that allows only SELECT/READ/FUNCTION (and blocks `load_extension`, `readfile`, ...), statement timeout, row cap
Prompts	`SECURITY_RULES` in every prompt	User text is wrapped as data, never as instructions
API	`middleware/`	JWT auth, per-IP rate limits (6/min, 60/day), login throttling, chats isolated per user and browser, generic 500s

LLM layer: Bedrock, model routing, structured outputs

- **Provider:** AWS Bedrock Converse API through `langchain-aws` (`ChatBedrockConverse`), region `us-west-2`, authenticated with a Bedrock API key (`AWS_BEARER_TOKEN_BEDROCK`). This is the same setup as the intelligence-assistant. Default model: **Claude Haiku 5.5** (`us.anthropic.claude-haiku-5-5`) for both tiers.
- **Model routing** (`llm_client.py`): nodes request a model by *pipeline step*, never by name. The order of precedence is: 1. a `MODEL_ROUTES` pin from `.env`; 2. retries (attempt > 1), which always escalate to the main model; 3. `sql_generation`, which is routed by plan complexity (fast for simple single-table plans, main otherwise); 4. otherwise, the step's default tier.

Swapping a tier or a single step to Sonnet is a one-line `.env` change. - **Structured outputs:** every non-prose step returns a Pydantic model (`llm_schemas.py`) through Bedrock's native JSON-schema mode. A reply that does not parse is retried once with tool calling. The code never uses regexes on model output. - **Observability:** every call is recorded as an `LLMCall` (step, model, routing reason, latency, input and output tokens, USD cost). These records drive the developer view, the usage table and the run history. - **Typical cost:** a full question takes 5 LLM calls, about 16k input / 1.5k output tokens, and **≈ $0.0025**. A refusal takes 1 call and ≈ $0.0002.

Step	Default tier	Prompt
intent_detection	fast	prompts.md#intent_detection
query_planner	main	prompts.md#query_planner
sql_generation	by complexity	prompts.md#sql_generation
sql_repair / sql_debugging / sql_optimization	main	prompts.md
generate_explanation / suggest_follow_ups / schema_answer	fast	prompts.md

Sample database

Summit Supply Co., a fictional US omnichannel retailer, stored in SQLite with 5 tables and deterministic synthetic data (fixed seed):

Table	Rows	Holds
`Employees`	333	Workforce, with a self-referencing `ManagerID` hierarchy
`Customers`	3,000	Consumers and businesses, with an account manager
`Products`	111	Catalog with price, cost, stock, launch and discontinuation
`Orders`	16,613	Order header: customer, sales rep, status, payment, shipping
`OrderItems`	30,449	Order lines at the historical price and discount

The data is shaped to make questions interesting: seasonality, churn, never-ordered customers and never-sold products, status driven by order age, overdue invoices. The full schema, ER diagram, every column's meaning and values, metrics and glossary are in **docs/database.md**.

API reference

All routes are under `/api`. Every route except `/auth/login`, `/auth/accounts` and `/health` needs `Authorization: Bearer <token>`. Chat routes also need `X-Client-Id`, which the browser generates.

Method	Path	Purpose
POST	`/auth/login`	`{username, password}` → `{token, expires_at, user}`

Method	Path	Purpose
GET	`/auth/me` · `/auth/accounts`	Current user · the two test accounts (names and roles only)
POST	`/chat`	`{question, thread_id?, execute?}` → full response
POST	`/chat/stream`	Same, as Server-Sent Events: `start`, `node` (×N), `token` (×N), `response`
GET / PATCH / DELETE	`/threads`, `/threads/{id}`	List, load, rename, delete chats
GET	`/export/{request_id}.csv`	Full CSV of an answer (re-runs its validated SQL, ≤ 50,000 rows)
GET	`/schema` · `/schema/graph`	Schema browser data · knowledge graph
GET	`/dev/llm` · `/dev/prompts` · `/dev/knowledge-graph` · `/dev/runs`	Developer only
GET	`/health`	Database, model and auth configuration status

A response's `type` is one of `sql`, `answer`, `clarification`, `refusal` or `error`. An `sql` response contains `sql`, `explanation`, `results`, `cost`, `validation`, `dialects`, `assumptions`, `mappings`, `suggestions` and `pipeline`, plus `dev` for the developer account.

Frontend

React 19 + TypeScript 5.9 + Vite 8 + Tailwind CSS 4. The look and feel follows the intelligence-assistant (gradient sidebar, pill composer, shimmer waiting state, "Dive deeper"), in an indigo theme.

```
frontend/src/
├── api/          client.ts (typed REST client), sse.ts (stream parser), types.ts (API contract)
├── context/      AuthContext (session, role), SettingsContext (theme, execute toggle)
├── hooks/        useChat (streaming turn state), useThreads, useSchema, useMediaQuery
├── lib/          sqlHighlight, download, format, pipeline (node → caption), storage, cn
├── types/        chat.ts (UI message model)
└── components/
    ├── ui/         Button, IconButton, Badge, Tabs, Switch, Spinner, Modal/Drawer, Toast, CopyButton, Markdown, Section
    ├── layout/     AppShell, Sidebar, ThreadItem, ProfileMenu
    ├── chat/       ChatView, MessageList, UserMessage, AssistantMessage, StreamingProgress, Composer, EmptyState, DiveDeeper
    ├── inspector/  Inspector + SqlTab, ExplanationTab, ResultsTab, PipelineTab, DevTab, RunsTab
    ├── sql/        SqlCode, ValidationChecklist, CostCard
    ├── schema/     SchemaDrawer
    ├── dev/        PromptsDialog
    └── auth/       LoginPage
```

Components are presentational. Data access lives in `api/` and `hooks/`, and cross-cutting state lives in `context/`. Only `VITE_*` variables reach the browser, so the frontend `.env` never holds secrets.

Configuration

Backend settings come from `backend/.env` (see `backend/.env.example` and `utils/config.py`):

Group	Variables
LLM (Bedrock)	`AWS_BEARER_TOKEN_BEDROCK`, `AWS_REGION`, `LLM_MODEL_FAST`, `LLM_MODEL_MAIN`, `LLM_MAX_TOKENS`, `LLM_INPUT_COST_PER_M`, `LLM_OUTPUT_COST_PER_M`
Routing	`MODEL_ROUTES` (JSON, e.g. `{"sql_generation": "us.anthropic.claude-sonnet-5-5"}`), `SQL_FAST_MODEL_MAX_COMPLEXITY`
Accounts	`AUTH_NORMAL_USERNAME/PASSWORD/DISPLAY_NAME`, `AUTH_DEV_USERNAME/PASSWORD/DISPLAY_NAME`, `AUTH_SECRET`, `AUTH_TOKEN_TTL_HOURS`
Guardrails	`MAX_RESULT_ROWS` (500), `QUERY_TIMEOUT_S` (5), `MAX_SQL_FIX_ATTEMPTS` (3), `EXPORT_MAX_ROWS` (50,000)
Features	`SUGGESTIONS_ENABLED`
API protection	`RATE_LIMIT_PER_MINUTE` (6), `RATE_LIMIT_PER_DAY` (60), `LOGIN_ATTEMPTS_PER_MINUTE` (10)

Frontend settings (`frontend/.env`): `VITE_API_BASE_URL` (empty when served by the backend) and `VITE_DEV_PROXY_TARGET`.

Project structure

```
.
├── backend/
│   ├── main.py                    FastAPI app: lifespan, CORS, error handlers, request log, SPA serving
│   ├── agent/
│   │   ├── supervisor/            graph build (supervisor.py), state, routing, traced() wrapper, dependencies
│   │   ├── intent/                receive_request, intent_detection, scope_validation
│   │   ├── retrieval/             schema_retrieval + the retrieval swarm
│   │   ├── planner/               query_planner, plan_grounding
│   │   ├── exec/                  sql_generation, sql_repair, sql_debugging, sql_optimization, execute_query,
│   │   │                          sqlite_runner (read-only DB), cost_estimator
│   │   ├── guardrails/            input_guard, sql_validator, sql_validation
│   │   ├── narrative/             generate_explanation, suggest_follow_ups, ask_clarification, schema_answer, return_response
│   │   ├── knowledge/             kb_resolver (phrase matching, join planning), us_geography
│   │   ├── llm_client.py          Bedrock gateway, model router, call records
│   │   └── llm_schemas.py         Pydantic contracts for every structured LLM step
│   ├── prompt_library/            one versioned module per LLM node + shared rule blocks
│   ├── routes/ controllers/ middleware/   HTTP layer (thin routes → controllers; auth, rate limit, errors)
│   ├── utils/                     settings, chat/run store
│   ├── knowledge_base/            semantic_layer.yaml, examples.yaml, generated dictionary + graph
│   ├── data/                      schema.sql, company.db, company_sample.xlsx
│   ├── scripts/                   generate_data, build_knowledge, generate_docs, check_llm, ask, export_docs_pdf
│   └── tests/                     pytest suite + ScriptedLLM test double
├── frontend/                      React app (see Frontend)
└── docs/                          architecture, workflow, prompts, database, assumptions, images, pdf/
```

Testing and quality

```
cd backend
pytest -q              # 137 tests, a few seconds, no network
ruff check .
cd ../frontend
npm run typecheck
```

Suite	Covers
`test_data`	Database shape, FK integrity, realism properties
`test_knowledge`	Knowledge build validation, phrase matching, value lookup, join planning
`test_sql_validator`	Each validator check, with positive and negative cases
`test_guardrails`	Prompt injection, write-SQL detection, read-only runner, authorizer
`test_planning`	Grounding, model routing, cost estimation, every worked example validates and runs
`test_graph`	The full LangGraph on a scripted LLM: every route, repair loop, follow-ups, refusals, failures
`test_api`	Auth, roles, rate limits, ownership, streaming, export

The graph tests use `ScriptedLLM`, which replays queued responses per step while using the **real** model router. They are fast and deterministic, and they also cover routing decisions. Live behaviour on Bedrock is checked with `scripts/check_llm.py` and `scripts/ask.py`.

Documentation index

Document	Contents
docs/architecture.md	System diagram, request sequence, knowledge pipeline, security, storage, observability
docs/workflow.md	The LangGraph: compiled graph, state, every node, routing rules, walkthroughs
docs/prompts.md	Every prompt in full, with its node, version, model tier and output contract
docs/database.md	Schema, ER diagram, column meanings and values, metrics, glossary
docs/assumptions.md	Assumptions and design decisions
`docs/pdf/`	PDF versions of all of the above and this README

Limitations and next steps

- **SQLite only for execution.** PostgreSQL and MySQL versions are translated (sqlglot) but not executed. Pointing the runner at another engine means swapping `ReadOnlyDatabase` and the `EXPLAIN` -based cost estimator.
- **One database per deployment.** The knowledge base is built per schema; a new database needs its semantic layer written.
- **In-memory rate limiter.** Fine for a single instance; use Redis for horizontal scaling.
- **No accuracy benchmark.** A golden-question evaluation set was deliberately left out of scope for now.
- **Docker and deployment** are the next step.